

Functions as arguments

CSCI
134

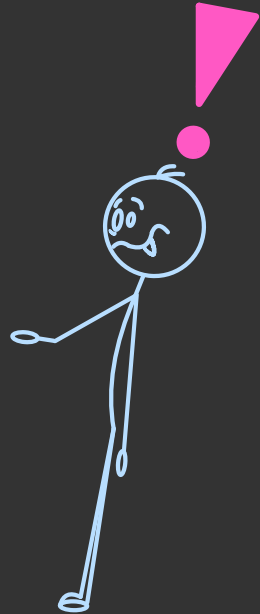
what does this **code** do?

```
In [1]: def square(num):  
...:     return pow(num, 2)  
...:  
  
In [2]: def do_it_twice(func, arg):  
...:     return func(func(arg))  
...:  
  
In [3]: do_it_twice(square, 3)  
Out[3]: ???
```



what does this **code** do?

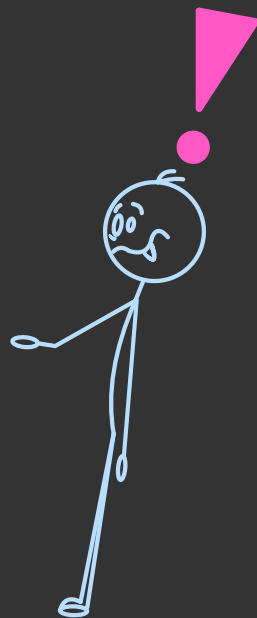
```
In [1]: def square(num):  
...:     return pow(num, 2)  
...:  
  
In [2]: def do_it_twice(func, arg):  
...:     return func(func(arg))  
...:  
  
In [3]: do_it_twice(square, 3)  
Out[3]: 81
```



like any expression, functions can be
arguments to other functions

```
In [1]: def square(num):  
...:     return pow(num, 2)  
...:  
  
In [2]: def do_it_twice(func, arg):  
...:     return func(func(arg))  
...:  
  
In [3]: do_it_twice(square, 3)  
Out[3]: 81
```

↙ a function!



exercise :

fill in the
blanks

```
In [1]: def do_it_twice(func, arg):  
...:     return func(func(arg))  
...:
```

```
In [2]: def square(num):  
...:     return pow(num, 2)  
...:
```

```
In [3]: def add_three(num):  
...:     return num + 3  
...:
```

```
In [4]: do_it_twice(square, 3)  
Out[4]: 81
```

```
In [5]: do_it_twice(add_three, 4)  
Out[5]: ?
```

```
In [6]: do_it_twice(type, 34.5)  
Out[6]: ?
```

this computes

`add_three(add_three(4))`

`func`

`func`

`arg`



```
In [1]: def do_it_twice(func, arg):  
...:     return func(func(arg))  
...:
```

```
In [2]: def square(num):  
...:     return pow(num, 2)  
...:
```

```
In [3]: def add_three(num):  
...:     return num + 3  
...:
```

```
In [4]: do_it_twice(square, 3)  
Out[4]: 81
```

```
In [5]: do_it_twice(add_three, 4)  
Out[5]: 10
```

```
In [6]: do_it_twice(type, 34.5)  
Out[6]: ?
```

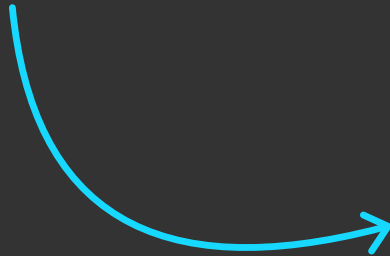
this computes

```
type(type(34.5))
```

func

func

arg



```
In [1]: def do_it_twice(func, arg):  
...:     return func(func(arg))  
...:
```

```
In [2]: def square(num):  
...:     return pow(num, 2)  
...:
```

```
In [3]: def add_three(num):  
...:     return num + 3  
...:
```

```
In [4]: do_it_twice(square, 3)  
Out[4]: 81
```

```
In [5]: do_it_twice(add_three, 4)  
Out[5]: 10
```

```
In [6]: do_it_twice(type, 34.5)  
Out[6]: type
```